

# Load Balancing in Cloud Environment to Minimize Average Response Time

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**Abstract**— Cloud computing is a recent and fastest-growing technology. Load balancing is a major challenge in cloud where the resources have to be directed to their respective servers so that the whole system works efficiently by distributing the workload efficiently. In this study, we compared the average response times of the six Load Balancing Algorithms, including Throttled, RR, ESET, Dynamic, Ip Hash, and Least Connection, to determine which one is the best. This research paper uses cloud analyst platform to perform a thorough comparative study of all six Load balancing algorithms along with three Service Broker Policies to find out with which best results are achieved.

**Keywords**— *Load Balancing Policy (LBP), Service Broker Policy (SBP), Cloud Computing, Response Time (RT), Data Center (DC), Virtual Machine (VM), User Base (UB).*

## I. INTRODUCTION

The fastest-evolving technology in the IT sector, cloud computing is a new way to supply services that are paid for only as they are used. Cloud computing has become a well-known technology that has been embraced by both businesses and academics, offering a flexible and effective way to store and retrieve materials. Management of infrastructure is done by cloud service providers, the cloud environment. The major challenge is scheduling incoming requests so that there is a minimum response time, effective resource use, and at the same time no resource is being underutilized. The process of load balancing involves dividing workloads among several computer resources in order to maximize resource utilization and guarantee quick and dependable service delivery. The average response time, or the time to process a request by the system and for the user to receive the response, is one of the crucial elements influencing the quality of cloud services. Poor user experience, lower customer satisfaction, and even financial loss can result from slow response times. The concept of

Load Balancing Algorithm can be used in a number of fields, including ANN and function optimization. They can also be used to arrange a simple workflow in virtual computers running on the cloud.

This paper provides an insight about various Load Balancing Algorithms in the cloud computing environment. We have used the Cloud Analyst simulator for the implementation and comparative analysis of those algorithms using various policies of service broker.

## II. LITERATURE REVIEW

Nguyen et al., in their paper, presented a custom throttled algorithm which would upgrade load balancing, decrease requests that are in queue and improve processing time and response times of cloud in cloud analyst environment. This algorithm limits the number of queued requests for delivery. When compared to the other two methods, this improves processing time and response time. One drawback of this approach is that it is complex, hence there is scope of improvement. [1]

This paper is basically representing a classification between several Load Balancing Algorithms, strategies and methods. It also discusses current approaches that are used in load balancing. This paper highlights the advantages and limitations of current approaches. It also discusses various ways or insights to improve the current models. This model lacks elasticity. It has used less resources. [2]

In this paper, the performance of the Load Balancing Algorithms is evaluated and given. Advantages include improvement in response time and minimization in cost of execution which ensures the respect of multi-class SLAs.

One drawback of this model would be its complex structure. [3]

Karadawy et al has done thorough analysis in cloud analyst, simulating load balancing using various algorithms using various SBPs. Basically three algorithms are discussed in detail, namely Round Robin, Throttled and Equally Spread Current Execution. It offers the best avg. RT for closest Data Center (DC) strategies, but it excludes a virtualized cloud environment from DC execution times. [4]

In this paper, various load balancing techniques using various SBPs is simulated. Also a custom made dynamic Load Balancing Algorithm is developed for integration of characteristics of other policies. It gives the best result among all parameters but doesn't have the least response time. [5]

This paper is an analysis of how good the current load balancing is, in respect to execution. This analysis also included comparison of various SBPs on cloud analyst. Using these criteria performance evaluation is done. RTs for LBPs based on geographic region are not shown in this article. [6]

This paper displays the analysis done using ESCE and Throttled algorithms to assess the accuracy of these algorithms. Detailed analysis using avg. RT, service time for DC and overall cost is done. Improvement of overall RT with reduced cost is required. [7]

In [8] Suhuna et al. have studied different dynamic load balancing algorithm and analyzed their performance.

## Load balancing Algorithms

### A. Round Robin

Round robin is a preemptive scheduling algorithm. It follows the first come first serve rule except it has a time quantum or time slice (generally 10 to 100 milliseconds) which lets a job to execute for that amount of time only. The ready queue is treated as a circular queue. The job is assigned to a server up to 1 time quantum. When the server completes one round execution of the job it checks which jobs are left to complete their execution and then it repeats the same process till all the jobs complete their execution. This algorithm does not choose the load based on the server load, it chooses them randomly.

### B. Equally Spread Current Execution

This Load Balancing Algorithm is a method to distribute computing resources among several VMs or containers. It works by monitoring the current execution load on each of the available resources and assigning new requests or tasks to the resource with the lowest current execution load. However, this algorithm may not take into account the specific requirements of individual applications or the availability of specialized hardware or software resources, so it may be necessary to use other load balancing techniques in more complex cloud computing environments.

### C. Throttled

A. This algorithm is based on the availability of VMs. A configuration table of VMs and their status is made. If a job is very well suited to a particular VM and it is available then the load balancer assigns that job to the VM. If no VMs are available then the load balancer puts the job in queue and waits until a preferred VM is found. This algorithm thus uses the resources effectively. The disadvantage of this algorithm is that it needs to scan all the VMs, if the required VM is at the bottom of the list. It does not consider the current load of the VM.

### D. IpHash

This algorithm is a kind of hashing algorithm which distributes network traffic among various servers or devices. It passes IP address of source in the hash function to obtain a hash value. This hash value is mapped to an available server or device using predefined mapping algorithms. After the source is mapped to its server, flow of data begins. This is not the best algorithm but it makes sure that the data transfer is safe [9]. It ensures the flow of data from source to server seamlessly. Also this algorithm can handle load balancing traffic of multiple DCs. It is not better than Round-Robin or least connections algorithm for some specific situations.

### E. Least Connection

Using the Least Connection Load Balancing Algorithm, multiple servers or other devices can share incoming network traffic in computer networking. It operates by keeping track of each server's number of active connections and allocating new connections to the server with the fewest active connections. This makes it possible to prevent any server from becoming overburdened with active connections and to disperse the load among all accessible servers in an equitable manner. Applications with persistent connections or irregular traffic patterns benefit the most from the Least Connection algorithm. As the number of connections may fluctuate quickly and cause an uneven load distribution it might not be appropriate for applications with short-lived connections [10].

### F. Dynamic Load Balancing Algorithm

Static algorithms are more rigid than dynamic algorithms, which depend on the system's current state rather than prior information. Load balancing through dynamic means is made possible due to a distributed system in two different ways: distributed and non-distributed. In the distributed model, this method is carried out by each node in the system, and the load-balancing work is distributed among these servers. Cooperative and non-cooperative interactions between nodes can be used to achieve load balancing [11]. In the first, the nodes collaborate to enhance the overall response time, among other things, by working side by side to reach the goal state. In the second type, every node operates independently to achieve a local objective.

**Table I**  
**Comparison of Load Balancing Algorithms**

| Algorithms                             | Pros                                                                                                                                                      | Cons                                                                                                                                                                                                                         |
|----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Round Robin(RR)                        | It uses a very simple model which is easy to understand and implement. Average response time is improved.                                                 | It doesn't check the current load of the server. Takes huge time in Context Switching.                                                                                                                                       |
| Equally Spread Current Execution(ESCE) | It assigns the task to the VMs according to their priorities. Task is evenly distributed among all the VMs.                                               | It doesn't take into account the specific requirements of individual applications or the availability of specialized hardware or software resources.                                                                         |
| Throttled                              | It assigns tasks on the basis of the availability of the VM and the best suit for the task.                                                               | It doesn't check the current load of the server.                                                                                                                                                                             |
| IpHash                                 | It enables the constant routing of all requests from a certain source IP address to a single server. This preserves session affinity.                     | It may cause an imbalance in the distribution of traffic when the source IP addresses are not spread equally across the network. Some servers could have unusually high traffic levels, while others might be underutilized. |
| Least Connection(LC)                   | It can optimize response times and reduce the possibility of server overload by sending new connections to the server with the fewest active connections. | This might make the load balancing solution more difficult and could necessitate more monitoring and administration work.                                                                                                    |

|         |                                                                                                                             |                                                        |
|---------|-----------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------|
| Dynamic | It increases performance and efficiency of parallel programming. Minimize execution time and maximize resource utilization. | It uses a very complex model which increases overhead. |
|---------|-----------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------|

### III. SERVICE BROKER POLICIES

When a user makes a request, the SBPs help in deciding which DC should process the user's request. Service Brokers acts as a mediator between the user and the data center (customer and provider). Thus they are in charge of managing the traffic flow between User Bases and DCs. It also helps DCs by relieving them of acquiring users and enables integrated access to multiple cloud services.

There are three types of SBPs which are already present in the Cloud Analyst :

1. Closest DC (CDC)
2. Optimise Response Time Policy(ORT)
3. Dynamically reconfigurable routing with load balancing.(DRR)

#### A. CDC Policy

This policy selects the DC which is nearest to the user. It offers a data center with the smallest possible network latency. A DC is selected at random if there are many DCs in the area.

#### B. ORT Policy:

It first uses the previous policy to identify the closest DC, but as soon as that DC's performance (which takes response time into account) begins to decline, it evaluates each DC's current response time and searches for the one with the shortest estimated response time. However, it's possible that the nearest and quickest DC was chosen at random.

#### C. DRR with load balancing.

This policy is an addition to the nearest DC policy, with the exception that it also considers the load that the VMs are currently under. After selecting the DC which is closest to the user, it checks if it can process more load. Its responsibility is to scale the deployment of the application in response to the load that it is experiencing. Accordingly, it either raises or reduces the number of VMs. It also takes into account the current processing time. This policy is under development.

#### IV. DEPLOYMENT CONFIGURATION

In this section, we are going to discuss how we performed simulation on the Cloud Analyst. By adjusting the configurations of the various Cloud Analyst tool components, we will now assess the various load balancing policies. As shown in the Figures given below, we have set the settings for the deployment of the application, UB configuration, DC configuration, SBP and LBP, respectively. We have identified the locations of 619 user bases across six different geographical areas. To handle the demands from 619 users, we used five DCs. Each of the five DCs contains a total of 25 VMs. We chose the RR algorithm as our load balancing policy. The simulation will last for 60 minutes. Similarly, the simulation is done for the other five load balancing algorithm(Equally Spread Current Execution, Throttled, IpHash, Least Connection and dynamic load balancing algorithm) The modeller can run the simulation multiple times with varying parameter values thanks to cloud analyst. These figures simulation results can be more effectively and simply seen in graphical form.

#### V. RESULTS AND FINDING

In this section, we have attached the results from the simulation of dynamic load balancing(DLB) algorithm based on 3 service broker policies. Fig 1 represents the simulation result for DLB using closest data center policy. Fig 2 and 3 shows the User Base Hourly Response Times and Cost respectively for the same. Fig 4 represents the simulation result for DLB using optimize response time policy. Fig 5 and 6 shows the User Base Hourly Response Times and Cost respectively for the same. Fig 7 represents the simulation result for DLB using Dynamically re-configurable routing policy. Fig 8 and 9 shows the User Base Hourly Response Times and Cost respectively for the same.

##### A. Dynamic Load Balancing Algorithm with Closest DC



Fig. 1. Simulation Result(CDC)

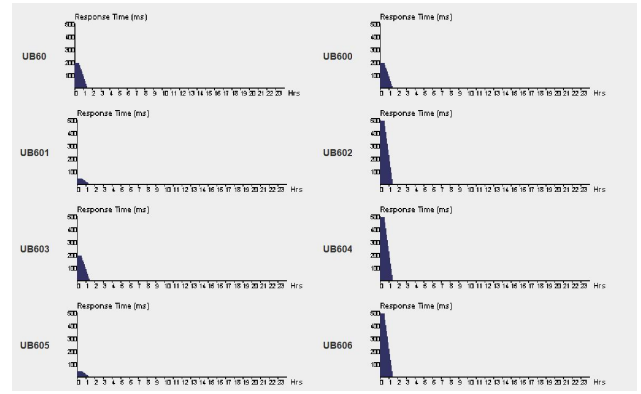


Fig. 2. User Base Hourly Response Times(CDC)

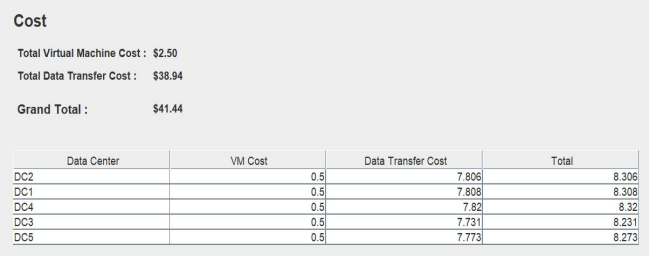


Fig. 3. Cost(CDC)

##### B. Proposed Dynamic Load Balancing Algorithm with Optimize response time



Fig. 4. Simulation Result(ORT)

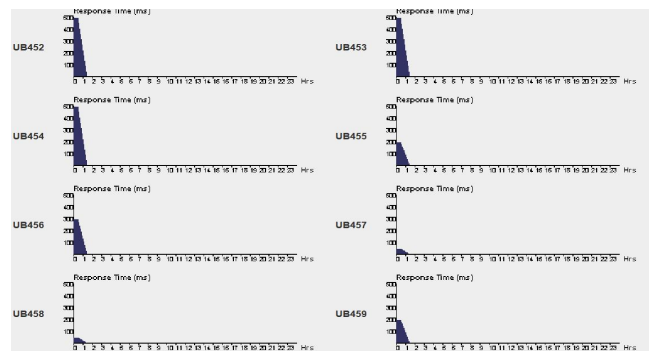


Fig. 5. User Base Hourly Response Times(ORT)

| <b>Cost</b>                         |         |                    |       |  |
|-------------------------------------|---------|--------------------|-------|--|
| Total Virtual Machine Cost : \$2.50 |         |                    |       |  |
| Total Data Transfer Cost : \$38.94  |         |                    |       |  |
| <b>Grand Total : \$41.44</b>        |         |                    |       |  |
| Data Center                         | VM Cost | Data Transfer Cost | Total |  |
| DC2                                 | 0.5     | 7.758              | 8.258 |  |
| DC1                                 | 0.5     | 7.907              | 8.407 |  |
| DC4                                 | 0.5     | 7.704              | 8.204 |  |
| DC3                                 | 0.5     | 7.793              | 8.293 |  |
| DC5                                 | 0.5     | 7.776              | 8.276 |  |

Fig. 6. Cost(ORT)

### C. Dynamic Load Balancing Algorithm with Dynamically reconfigurable routing

Overall Response Time Summary

|                                   |              |              |
|-----------------------------------|--------------|--------------|
| Average (ms)                      | Minimum (ms) | Maximum (ms) |
| Overall Response Time: 287.76     | 36.61        | 877.26       |
| Data Center Processing Time: 0.92 | 0.01         | 459.00       |

Export Results

Response Time By Region

| Userbase | Avg (ms) | Min (ms) | Max (ms) |
|----------|----------|----------|----------|
| UB100    | 50.694   | 39.056   | 61.695   |
| UB101    | 200.41   | 161.969  | 240.021  |
| UB102    | 501.336  | 396.187  | 588.671  |
| UB103    | 502.293  | 405.084  | 608.36   |
| UB104    | 497.784  | 382.967  | 600.781  |
| UB105    | 50.839   | 40.507   | 64.318   |
| UB106    | 200.256  | 163.364  | 238.661  |
| UB107    | 501.051  | 387.773  | 598.665  |
| UB108    | 501.346  | 395.302  | 605.313  |
| UB109    | 499.864  | 401.096  | 608.825  |
| UB110    | 500.921  | 388.074  | 605.525  |
| UB111    | 50.867   | 40.705   | 65.613   |
| UB112    | 201.356  | 158.779  | 243.517  |
| UB113    | 502.209  | 400.107  | 609.031  |
| UB114    | 502.488  | 383.515  | 641.972  |
| UB115    | 50.65    | 40.486   | 61.239   |
| UB116    | 201.499  | 158.351  | 246.931  |
| UB117    | 499.408  | 403.181  | 595.249  |
| UB118    | 500.008  | 360.356  | 630.261  |
| UB119    | 200.219  | 149.256  | 254.309  |
| UB120    | 300.368  | 245.704  | 360.968  |

Fig. 7. Simulation Result(DRR)

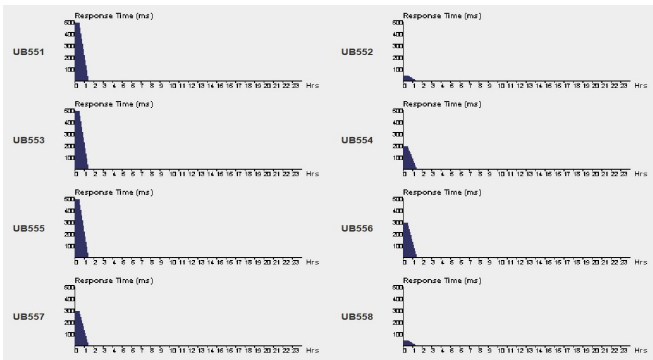


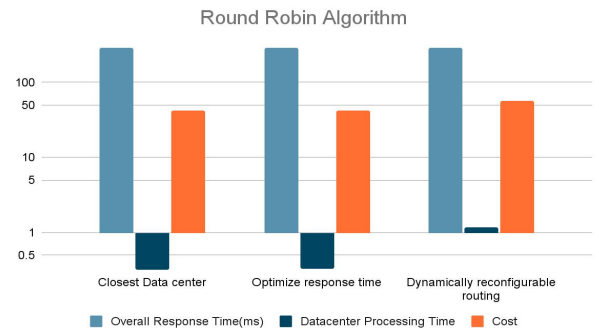
Fig. 8. User Base Hourly Response Times(DRR)

| <b>Cost</b>                          |         |                    |        |  |
|--------------------------------------|---------|--------------------|--------|--|
| Total Virtual Machine Cost : \$16.45 |         |                    |        |  |
| Total Data Transfer Cost : \$38.94   |         |                    |        |  |
| <b>Grand Total : \$55.39</b>         |         |                    |        |  |
| Data Center                          | VM Cost | Data Transfer Cost | Total  |  |
| DC2                                  | 3.162   | 7.773              | 10.935 |  |
| DC1                                  | 3.255   | 7.813              | 11.068 |  |
| DC4                                  | 3.328   | 7.767              | 11.096 |  |
| DC3                                  | 3.352   | 7.795              | 11.147 |  |
| DC5                                  | 3.352   | 7.789              | 11.141 |  |

Fig. 9. Cost(DRR)

## VI. COMPARISON OF LOAD BALANCING ALGORITHM AND RESULTS

This section analyzes the six Load Balancing Algorithms (RR, ESCE, IpHash, Throttled, DLB, LC) with the three Service Broker Policies(CDC, ORT, DRR) on the basis of parameters(Overall RT, DC processing time, cost)and shown in fig-10,11,12,13,14 &15 respectively.



A.

B. Fig. 10. Analysis of the performance parameters of RR based on 3 SBPs.

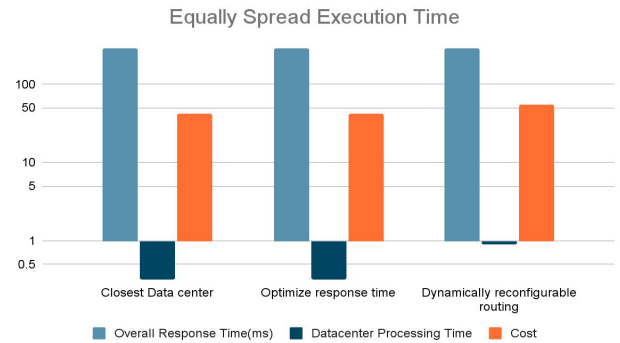


Fig. 11. Analysis of the performance parameters of ESET based on 3 SBPs.

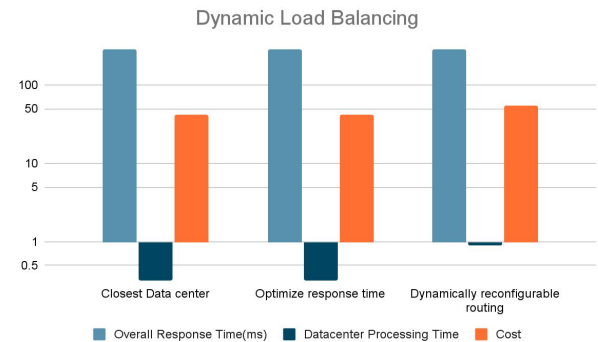


Fig. 12. Analysis of the performance parameters of DLB based on 3 SBPs.

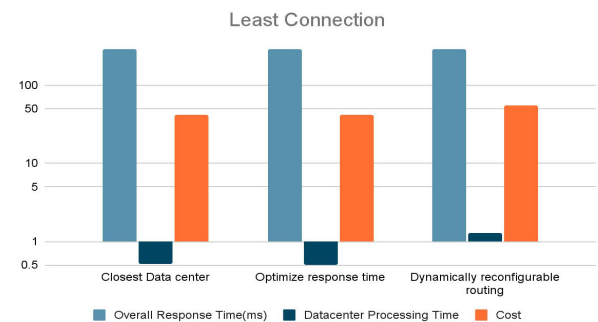


Fig. 13. Analysis of the performance parameters of Least Connection based on 3 SBPs.

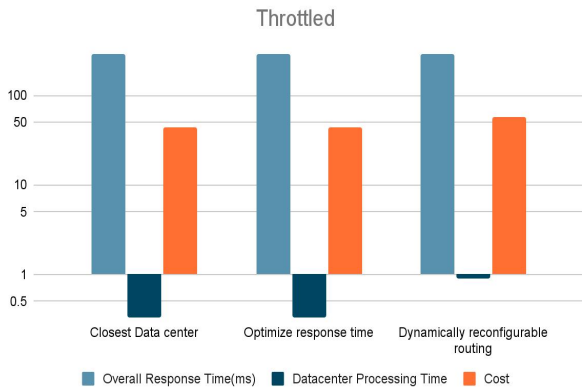
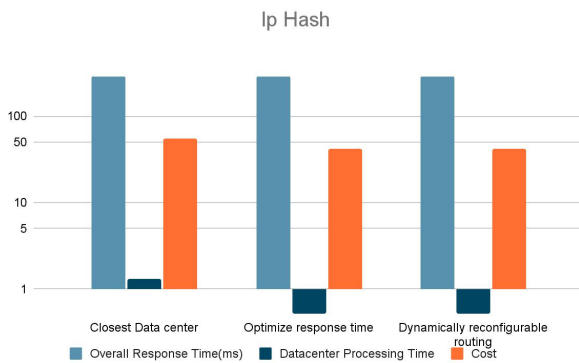


Fig.

14. Analysis of the performance parameters of Throttled based on 3 SBPs.



C. Fig. 15. Analysis of the performance parameters of IP Hash based on 3SBPs.

## VII. CONCLUSION

Comparing all the simulation results of the Load Balancing Algorithms, we can conclude that on the basis of Response Time, the Dynamic Load Balancing Algorithm with the Optimize Response Time SBP is the most efficient one. By dynamically assigning resources based on their availability and consumption, dynamic Load Balancing Algorithms, like the one used in the provided code, have the advantage of being able to respond to changing workload needs. This improves resource utilization and can potentially lower expenses. However, Load Balancing Algorithms can also increase system complexity and overhead, particularly in large-scale and distributed environments. In order to make sure that Load Balancing Algorithms are effective, efficient, and scalable, it is crucial to thoroughly design and assess them. Future Scope for this topic involves Machine learning-based load balancing, Load balancing for multi-cloud environments, Load balancing for edge computing, Load balancing for heterogeneous cloud environments.

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